UF UNIVERSITY of FLORIDA

Department of Mathematics

MAP 2302
Exam 3A
Spring 2022

A. Sign your bubble sheet on the back at the bottom in ink.

B. In pencil, write and encode in the spaces indicated:

1) Name (last name, first initial, middle initial)

2) UF ID number

3) Section number

C. Under "special codes" code in the test ID numbers 3, 1.

1	2	●	4	5	6	7	8	9	0
●	2	3	4	5	6	7	8	9	0

D. At the top right of your answer sheet, for "Test Form Code", encode A.

● B C D E

E. 1) This test consists of 12 multiple choice questions of five points in value, plus 4 free response questions worth 40 points.

2) The time allowed is 100 minutes.

3) You may write on the test.

4) Raise your hand if you need more scratch paper or if you have a problem with your test. DO NOT LEAVE YOUR SEAT UNLESS YOU ARE FINISHED WITH THE TEST.

F. KEEP YOUR BUBBLE SHEET COVERED AT ALL TIMES.

G. When you are finished:

1) Before turning in your test **check carefully for transcribing errors**. Any mistakes you leave in are there to stay.

2) You must turn in your scantron and tearoff sheets to your discussion leader or exam proctor. Be prepared to show your picture I.D. with a legible signature.

3) The answers will be posted in Canvas within one day after the exam. Your discussion leader will return your tearoff sheet with your exam score in discussion. Your score will also be posted in Canvas within one week of the exam.

NOTE: Be sure to bubble the answers to questions 1−12 on your scantron.

Questions 1 − 12 are worth 5 points each.

1. Is $g(t)$ continuous, piecewise continuous, or neither on $[0, 5]$?

$$g(t) = \begin{cases} \dfrac{\sin(2t)}{t} & t \neq 0 \\ 1 & t = 0 \end{cases}$$

a. continuous b. piecewise continuous c. neither

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2. Find the Laplace transform $\mathscr{L}\{(t+1)^4\}$

a. $\dfrac{12}{s^5} + \dfrac{6}{s^4} + \dfrac{6}{s^3} + \dfrac{4}{s^2} + \dfrac{1}{s}$

b. $\dfrac{24}{s^5} + \dfrac{12}{s^4} + \dfrac{6}{s^3} + \dfrac{3}{s^2} + \dfrac{1}{s}$

c. $\dfrac{24}{s^5} + \dfrac{12}{s^4} + \dfrac{12}{s^3} + \dfrac{4}{s^2} + \dfrac{1}{s}$

d. $\dfrac{24}{s^5} + \dfrac{24}{s^4} + \dfrac{12}{s^3} + \dfrac{4}{s^2} + \dfrac{1}{s}$

e. $\dfrac{12}{s^5} + \dfrac{12}{s^4} + \dfrac{6}{s^3} + \dfrac{4}{s^2} + \dfrac{1}{s}$

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3. Find the inverse Laplace transform $\mathscr{L}^{-1}\{\dfrac{9}{(2s+3)^3}\}$

a. $\frac{9}{16}t^2e^{-\frac{3}{2}t}$ b. $\frac{1}{3}t^2e^{\frac{3}{2}t}$ c. $\frac{1}{3}t^2e^{-\frac{3}{2}t}$ d. $\frac{9}{16}t^2e^{\frac{3}{2}t}$ e. $\frac{7}{16}t^2e^{-\frac{3}{2}t}$

4. Solve for $Y(s)$, the Laplace transform of the solution $y(t)$ of the initial value problem.

$$y'' + 6\,y = 4t^2 - 3,\ y(0) = 0,\ y'(0) = -9$$

a. $Y(s) = \dfrac{-9s^3 - 3s^2 + 8}{s^3(s^2+6)}$

b. $Y(s) = \dfrac{-5s^3 - 3s^2 + 8}{s^3(s^2+6)}$

c. $Y(s) = \dfrac{-8s^3 - 4s^2 + 7}{s^3(s^2+9)}$

d. $Y(s) = \dfrac{-7s^3 - 3s^2 + 8}{s^3(s^2+3)}$

e. $Y(s) = \dfrac{-9s^3 + 3s^2 + 8}{s^3(s^2+6)}$

5., 6. A linear system is goverened by the given initial value problem

$$y'' - 8y' + 17y = g(t),\ y(0) = 0,\ y'(0) = 1$$

5. Find the transfer function $H(s)$.

a. $\dfrac{1}{(s-4)^2+1} + \dfrac{G(s)}{(s-4)^2+1}$

b. $\dfrac{1}{(s-4)^2+1}$

c. $\dfrac{s}{(s-4)^2+1} + \dfrac{G(s)}{(s-4)^2+1}$

d. $\dfrac{s-4}{(s-4)^2+1} + \dfrac{3}{(s-4)^2+1}$

e. $\dfrac{s-4}{(s-4)^2+1} + \dfrac{4G(s)}{(s-4)^2+1}$

6. Find the impulse response function $h(t)$.

a. $e^{4t}\cos t$

b. $e^{4t}\sin t$

c. $e^{4t}\cos t + \displaystyle\int_0^t e^{t-v}\sin(t-v)g(v)\,dv$

d. $e^{4t}\cos t + 3e^{4t}\sin t$

e. $e^{4t}\sin t + \displaystyle\int_0^t e^{t-v}\sin(t-v)g(v)\,dv$

7. Determine the inverse Laplace transform of $\dfrac{se^{-5s}}{s^2+10s+34}$.

a. $e^{-5(t+5)}\left[\cos(3(t+5)) - \frac{5}{3}\sin(3(t+5))\right]u(t+5)$

b. $e^{-5(t-5)}\left[\cos(3(t-5)) - \frac{5}{3}\sin(3(t-5))\right]u(t-5)$

c. $e^{-5(t-5)}\left[\cos(3(t-5)) + \frac{5}{3}\sin(3(t-5))\right]u(t+5)$

d. $e^{-5(t+5)}\left[\cos(3(t-5)) - \frac{5}{3}\sin(3(t-5))\right]u(t+5)$

e. $e^{-5(t-5)}\left[\cos(\sqrt{3}(t-5)) + \frac{5}{3}\sin(\sqrt{3}(t-5))\right]u(t-5)$

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8. Determine the inverse Laplace transform of the function below.

$$\frac{6s-102}{5s^2-20s+65}$$

a. $\frac{6}{5}e^{-2t}\cos(3t) - 4e^{-2t}\sin(3t)$

b. $\frac{3}{5}e^{-2t}\cos(3t) - 8e^{-2t}\sin(3t)$

c. $\frac{2}{5}e^{2t}\cos(3t) - 8e^{2t}\sin(3t)$

d. $\frac{4}{5}e^{2t}\cos(3t) - 6e^{2t}\sin(3t)$

e. $\frac{6}{5}e^{2t}\cos(3t) - 6e^{2t}\sin(3t)$

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9. Use the convolution theorem to obtain a formula for the solution to the given initial value problem where $g(t)$ is piecewise continuous on $[0,\infty)$ and of exponential order.

$$y'' + 9y = g(t),\ y(0) = 4,\ y'(0) = 0$$

a. $y(t) = 4\cos(3t) + \dfrac{1}{2}\displaystyle\int_0^t \sin(3(t-v))g(v)dv$

b. $y(t) = 4\cos(3t) + \dfrac{1}{3}\displaystyle\int_0^t \sin(3(t-v))g(v)dv$

c. $y(t) = 2\cos(3t) + \dfrac{1}{3}\displaystyle\int_0^t \sin(3(t-v))g(v)dv$

d. $y(t) = 2\cos(3t) + \dfrac{1}{2}\displaystyle\int_0^t \sin(3(t-v))g(v)dv$

e. $y(t) = 4\sin(3t) + \dfrac{1}{2}\displaystyle\int_0^t \sin(3(t-v))g(v)dv$

10. Which of the following functions are of exponential order?

I. $\sin(e^{t^3}) + t^{99}e^{42t}$ II. t^3 III. e^{t^2}
IV. $e^{\frac{t^2}{t+2}}$ V. $t \ln t$

a. I, III, IV b. I, II, IV,V c. II, IV, V d. II, III, V e. all of them

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11. Solve the integral equation $y(t) + 9\int_0^t (t-v)y(v)dv = 9t^2$.

a. $y(t) = 2(1 - \sin(3t))$

b. $y(t) = 2(t - \sin(3t))$

c. $y(t) = 2(t - \cos(3t))$

d. $y(t) = 2(1 - \cos(3t))$

e. $y(t) = 1 - 2\cos(3t)$

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12. Find the Laplace transform of $3e^t\delta(t-4)$

a. $3e^{4s}$ b $3e^{4(s-1)}$ c. $3e^{4(1-s)}$

d. $3e^{1-4s}$ e. $3e^{4s-1}$

Elementary Laplace Transforms

$f(t)$	$\mathcal{L}\{f(t)\}$
C	$\frac{C}{s}, s > 0$
t^n	$\frac{n!}{s^{n+1}}, s > 0$
e^{at}	$\frac{1}{s-a}, \quad s > a$
$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a,\ n = 0, 1, 2, \ldots$
$\sin(bt)$	$\frac{b}{s^2+b^2}, s > 0$
$\cos(bt)$	$\frac{s}{s^2+b^2}, s > 0$
$e^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}, \quad s > a$
$e^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}, \quad s > a$
$\sinh(bt)$	$\frac{b}{s^2-b^2}, s > \lvert b\rvert$
$\cosh(bt)$	$\frac{s}{s^2-b^2}, s > \lvert b\rvert$
$u(t-a)$	$\frac{e^{-as}}{s}, s > 0$
$\delta(t-a)$	e^{-as}

Properties of the Laplace transform

$\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$

$\mathcal{L}\{cf\} = c\mathcal{L}\{f\}$ for any constant c

$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a)$ where $F(s) = \mathcal{L}\{f\}(s)$

$\mathcal{L}\{f'\}(s) = sF(s) - f(0)$

$\mathcal{L}\{f''\}(s) = s^2F(s) - sf(0) - f'(0)$

$\mathcal{L}\{f^{(n)}\}(s) = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0)$

$\mathcal{L}\{t^nf(t)\} = (-1)^n\frac{d^nF}{ds^n}(s)$

$\mathcal{L}\{ty'\}(s) = -sY'(s) - Y(s)$ where $Y(s) = \mathcal{L}\{y\}(s)$

$\mathcal{L}\{ty''\}(s) = -s^2Y'(s) - 2sY(s) + y(0)$

MAP 2302 Exam 3A, Part II Free Response

Name: _____ Section #: _____

SHOW ALL WORK TO RECEIVE FULL CREDIT

1. (11 pts) Use the method of Laplace transforms to solve the initial value problem.

$$y'' + y = 3u(t-3),\ y(0) = 0,\ y'(0) = 1.$$

2. (11 pts) Solve the initial value problem

$$y'' + 7y' + 10y = \delta(t-2),\ y(0) = 0,\ y'(0) = 0$$

3. (10 pts) Use the convolution theorem to find the inverse Laplace transform of the function below. No credit will be granted for partial fraction decomposition solutions.

$$Y(s) = \frac{3}{s^2(s^2+9)}.$$

4. (8 pts) Find the Laplace transform of $g(t)$ and the restriction on s **using the definition of the Laplace transform. No credit will be granted for other methods or use of tables**.

$$g(t) = \begin{cases} 1 & t \leq 4 \\ t-2 & t > 4 \end{cases}$$

University of Florida Honor Pledge:

On my honor, I have neither given nor received unauthorized aid doing this exam.

Signature: _____